

DAVID VALAREZO

(201) 838-0168 | dev38@cornell.edu | [linkedin.com/in/davidvalarezo](https://www.linkedin.com/in/davidvalarezo) | github.com/dvala041 | dvala041.github.io/portfolio

EDUCATION

Cornell University, College of Engineering

B.S in Computer Science, Minor in Artificial Intelligence

GPA: 3.65

Ithaca, NY

Expected May 2027

Coursework: Deep Learning, High Performance Parallel Computing, Systems Programming, Machine Learning, Functional Programming, Algorithms, Object Oriented Programming and Data Structures

EXPERIENCE

Datadog

Software Engineering Intern

- Incoming summer 2026

New York, NY

Jun 2026 – Aug 2026

Capital One

Software Engineering Intern

- Built backend monitoring system with modular plugin architecture for café kiosk platform serving **30+** locations, increasing system uptime and reducing manual support tickets
- Reduced kiosk recovery time by **60%** ($5 \rightarrow 2$ minutes) by developing **12+** TypeScript plugins that automatically detect and restart failed hardware components
- Designed high-throughput REST API and optimized DynamoDB schema processing **60,000+** daily health checks and **6 million+** cashier check transactions, enabling **10+** support engineers to reduce incident response time

McLean, VA

Jun 2025 – Aug 2025

Cornell Hack4Impact

Tech Lead

- Led a team of 5 developers to build a member archive platform connecting **200+** alumni, overseeing sprint planning, code reviews, subteam meetings, and technical mentorship
- Architected full migration from MongoDB to Supabase, implementing new database schema and user auth
- Improved onboarding processes for **100+** applicants each semester, contributing to stronger recruiting outcomes

Ithaca, NY

Aug 2025 – Dec 2025

Cornell Bowers CIS ASCEND Program

LinkedIn Fellow

- 1 of 28 students (**14%** acceptance rate) from **200+** applicants chosen for LinkedIn-sponsored leadership program
- Led product design in team of 5 for AI-powered productivity tool by creating prototype for final presentation

Ithaca, NY

Jan 2025 – Apr 2025

PROJECTS

Distributed GEMM on Cerebras WSE-2 | *CSL, Cerebras WSE-2, Perlmutter*

March 2026 – Present

- Implementing distributed matrix multiplication on Cerebras WSE-2 using CSL, designing fabric routing for column broadcast and ring reduction across a 2D PE grid to maximize throughput on a wafer-scale AI accelerator

PatchTST | *PyTorch*

March 2026 – Present

- Reproducing PatchTST (ICLR 2023) in PyTorch, implementing patch-based tokenization, channel-independent encoding, and masked self-supervised pretraining to forecast weather data across a 512-step prediction horizon

Particle Simulation | *C++, MPI, Perlmutter*

Feb 2026 – Mar 2026

- Achieved 3.5x speedup scaling from $64 \rightarrow 256$ MPI processes on Perlmutter by implementing spatial binning, ghost regions, and non-blocking send/receive to minimize inter-process communication overhead across 2M particles

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, Golang, Ocaml, SQL, C++, C, HTML/CSS

Frameworks: CUDA, MPI, OpenMP, PyTorch, ReactJS, NodeJS, Prisma, Flask, FastAPI, JUnit

Libraries/Developer Tools: HuggingFace, Jenkins, NumPy, Postman, Git, Docker, DynamoDB, MongoDB, AWS